Pixelcoin: A Merge-Link Electronic Cash System for Peer-to-Peer Payments

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Abstract. Pixelcoin (PXL) is a next-generation decentralized cryptocurrency designed to deliver fast, low-cost, and scalable digital transactions. Built on the proprietary Merge-Link Blockchain, Pixelcoin introduces a novel block structure that integrates merge sort–based mathematical ordering directly into block validation. This architecture enhances transaction throughput, improves data integrity, and strengthens security without compromising decentralization. Pixelcoin operates using a Proof-of-Work (PoW) consensus mechanism while supporting smart contracts and decentralized applications (dApps). With a fixed supply and a community-driven vision, Pixelcoin aims to bridge the gap between practical digital payments, decentralized innovation, and memecoin culture.

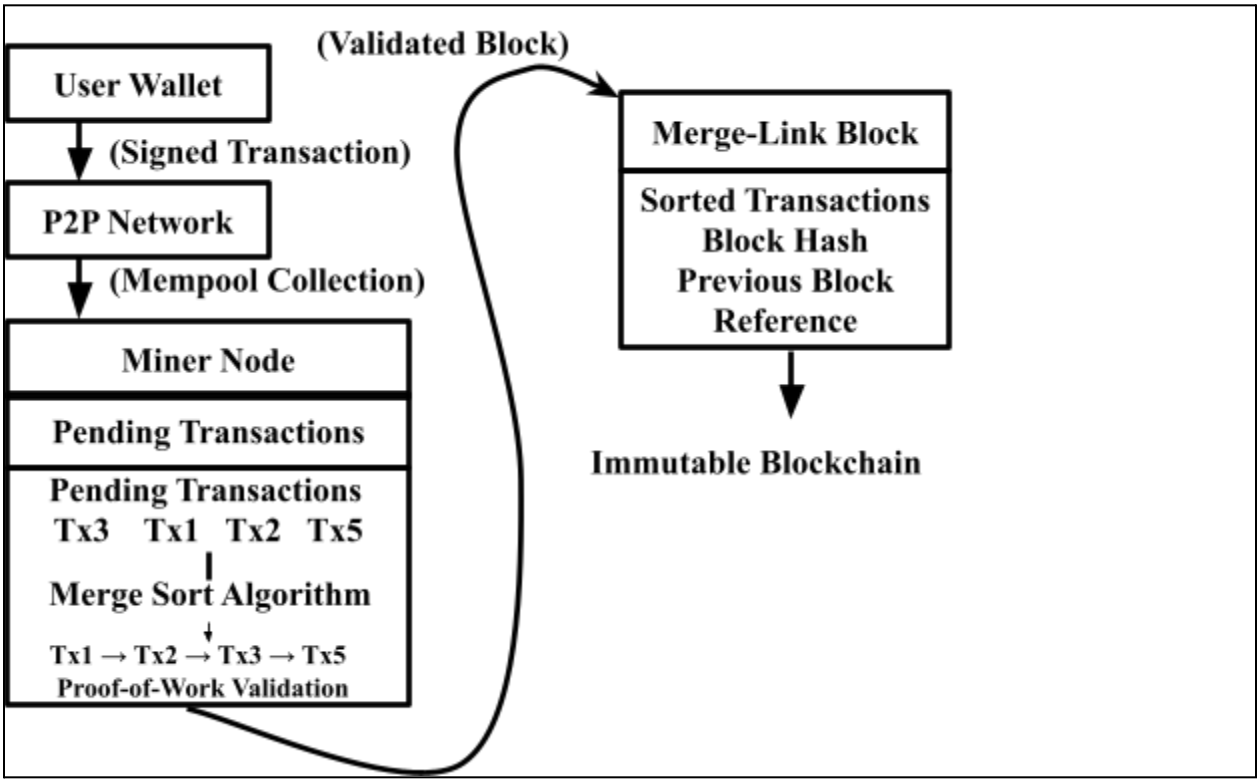
1. Introduction

Blockchain technology has transformed the way value is transferred, stored, and verified in the digital world. Since the introduction of Bitcoin, decentralized networks have demonstrated that trustless, peer-to-peer financial systems are possible without centralized authorities. However, as adoption has grown, so too have the limitations of first-generation blockchains. Slow confirmation times, high transaction fees, limited scalability, and inefficient data structures have created barriers to widespread everyday use. Many existing cryptocurrencies struggle to balance decentralization, security, and performance. Networks often become congested during periods of high activity, leading to delayed transactions and unpredictable fees. These challenges make traditional blockchains unsuitable for microtransactions, real-time payments, gaming economies, and socially driven financial use cases. As a result, there is a growing need for a blockchain architecture that is not only secure and decentralized, but also optimized for speed, efficiency, and scalability from the ground up. Pixelcoin (PXL) was created to address these challenges through a fundamentally redesigned blockchain framework known as the Merge-Link Blockchain. Rather than relying on purely sequential block data structures, Pixelcoin introduces mathematically ordered transaction processing by integrating merge sort algorithms directly into each block. This approach improves transaction organization, accelerates validation, and enhances resistance to manipulation while maintaining the core principles of decentralized consensus. Pixelcoin is designed to be both practical and accessible. With ultra-low transaction fees, fast confirmation times, and support for larger block sizes, it enables seamless peer-to-peer payments and microtransactions that are economically unfeasible on many legacy networks. At the same time, Pixelcoin functions as a flexible platform capable of supporting smart contracts and decentralized applications, allowing developers to build innovative financial, gaming, and digital ownership solutions on top of the network. Beyond its technical foundation, Pixelcoin embraces a community-driven philosophy. It is intended to function not only as a serious digital currency, but also as a memecoin with real utility—combining strong technology, open participation, and cultural appeal. This dual identity allows Pixelcoin to foster organic adoption while maintaining long-term value through a fixed supply and secure Proof-of-Work consensus mechanism. By merging advanced mathematical design, proven consensus security, and real-world

usability, Pixelcoin aims to represent the next step in blockchain evolution—one that moves beyond experimentation and into practical, everyday digital finance.

2. Transactions

Transactions are the core function of the Pixelcoin network. Pixelcoin is designed to process transactions quickly, securely, and at extremely low cost by leveraging the Merge-Link Blockchain architecture. Unlike traditional blockchains that process transactions in a loosely ordered list, Pixelcoin applies mathematical ordering and linking at the block level, improving efficiency and verification speed across the network.



Pixelcoin's transaction system is purpose-built for speed, reliability, and real-world usability. By integrating mathematical ordering directly into block construction, Pixelcoin eliminates many inefficiencies found in traditional blockchains. The result is a transaction layer capable of supporting everyday payments, microtransactions, decentralized applications, and future scalability—all while maintaining the security and decentralization that define blockchain technology.

3. What is Pixelcoin?

Pixelcoin (PXL) is a decentralized peer-to-peer cryptocurrency built on the Merge-Link Blockchain architecture. Unlike conventional blockchains that simply append blocks sequentially, Merge-Link blocks organize transaction data using merge sort logic, enabling:

- Faster transaction validation
- Improved block data integrity

- More efficient miner verification
- Reduced network congestion

Pixelcoin is designed for everyday usability while remaining powerful enough to support smart contracts and decentralized applications.

4. Merge-Link Blockchain Architecture

4.1 Concept Overview

The Merge-Link Blockchain embeds a merge sort algorithm into each block's structure. Transactions are mathematically ordered and linked during block creation rather than after validation.

This results in:

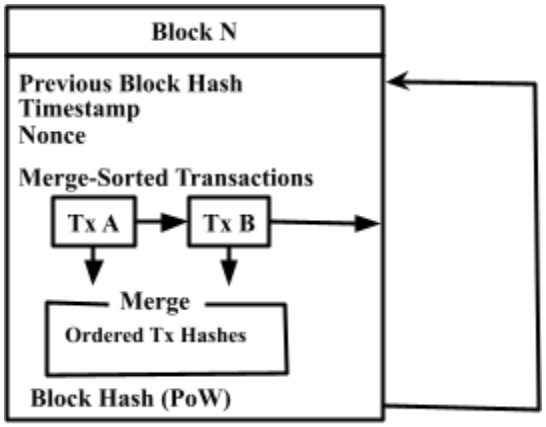
- Deterministic transaction ordering
- Reduced reorganization overhead
- Faster verification across nodes
- Increased resistance to manipulation

4.2 Block Structure

Each block contains:

- Block header (hash, timestamp, nonce)
- Sorted transaction tree (merge-linked)
- Previous block reference
- Proof-of-Work solution

4.3 Visual Representation:



Result: Faster validation, mathematically verifiable ordering, and efficient scaling.

5. Consensus Mechanism

Pixelcoin uses **Proof-of-Work (PoW)**, similar in concept to Bitcoin but adapted for the Merge-Link architecture.

Mining Process

1. Transactions are collected into a candidate block
2. Transactions are merge-sorted mathematically
3. Miners solve computational puzzles
4. Valid blocks are broadcast and verified
5. The block is permanently added to the chain

PoW ensures:

- Decentralization
- Network Security
- Resistance to censorship and tampering

6. Performance and Fees

Thanks to its optimized block structure:

- Larger block sizes are supported Higher throughput than legacy chains
- Transaction fees typically under $0.001 USD
- Fast confirmation times

This makes Pixelcoin ideal for microtransactions and real-world payments.

7. Smart Contracts and dApps

Pixelcoin is more than a currency—it is a **platform**.

Developers can build:

- Smart contracts
- Decentralized finance (DeFi) tools
- NFT platforms Blockchain games
- Social and creator-economy applications

The Merge-Link Blockchain ensures that smart contract execution remains efficient even as network usage grows.

8. Tokenomics

- **Token Name:** Pixelcoin
- **Ticker**: PXL
- **Total Supply**: 1,000,000,000 PXL
- **Circulating Supply (2025 est.)**: ~1 billion PXL
- **Supply Model**: Fixed (scarcity-driven)

The capped supply promotes long-term value preservation while supporting a sustainable mining ecosystem.

9. Use Cases

9.1 Digital Payments

Peer-to-peer transfers, merchant payments, and online purchases.

9.2 Microtransactions

Ideal for tipping, in-app purchases, and streaming rewards.

9.3 Social Impact

Low fees enable donations and rapid financial assistance to underserved communities.

9.4 Remittances

Fast, low-cost cross-border payments without intermediaries.

9.5 Smart Contracts & dApps

Supports DeFi, NFTs, gaming, and decentralized services.

9.6 Community & Memecoin Culture

Pixelcoin embraces community participation while maintaining real utility.

10. Development and Implementation

Pixelcoin was:

- **Fully developed in C++**
- Built as the **Merge-Link Blockchain library**
- Compiled using **CMake**
- Integrated into **Radiums Launchlab Token Program**

This foundation ensures performance, security, and long-term maintainability.

11. Vision & Conclusion

Pixelcoin represents a new evolution in blockchain design—combining mathematical precision, real-world usability, and community-driven culture. By addressing scalability and cost issues while supporting innovation through smart contracts and dApps, Pixelcoin positions itself as a compelling alternative to first-generation cryptocurrencies.

Whether you are a miner, developer, investor, or everyday user, **Pixelcoin offers a fast, efficient, and scalable future for digital finance.**